

PRAKTIKUM BASIS DATA

MODUL 11

NoSQL_Key_Value_Database



Oleh:

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**PROGRAM STUDI S1 REKAYASA
PERANGKAT LUNAK DIREKTORAT
KAMPUS PURWOKERTO
UNIVERSITAS TELKOM 2026**

MODUL 11

JURNAL

1. A.

```
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifine_0802
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE spotifine_0802;
cqlsh:spotifine_0802> |
```

B.

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifine_0802
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE spotifine_0802;
cqlsh:spotifine_0802> CREATE TABLE musician_data(
...     artist_id UUID PRIMARY KEY,
...     artist_name TEXT,
...     followers INT,
...     origin TEXT,
...     genres SET<TEXT>
... );
cqlsh:spotifine_0802> |
```

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing. Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotify_0802
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE spotify_0802;
cqlsh:spotify_0802> CREATE TABLE musician_data(
... artist_id UUID PRIMARY KEY,
... artist_name TEXT,
... followers INT,
... origin TEXT,
... genres SET<TEXT>
... );
cqlsh:spotify_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Dave Rodgers', 5000000, 'Italy', {'Eurobeat', 'Dance'});
cqlsh:spotify_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'YOASOBI', 25000000, 'Japan', {'J-Pop', 'Pop'});
cqlsh:spotify_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Manuel', 3000000, 'Italy', {'Eurobeat'});
cqlsh:spotify_0802>
```

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
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cqlsh> CREATE KEYSPACE spotify_0802
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE spotify_0802;
cqlsh:spotify_0802> CREATE TABLE musician_data(
... artist_id UUID PRIMARY KEY,
... artist_name TEXT,
... followers INT,
... origin TEXT,
... genres SET<TEXT>
... );
cqlsh:spotify_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Dave Rodgers', 5000000, 'Italy', {'Eurobeat', 'Dance'});
cqlsh:spotify_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'YOASOBI', 25000000, 'Japan', {'J-Pop', 'Pop'});
cqlsh:spotify_0802> INSERT INTO musician_data(artist_id,artist_name,followers,origin,genres)
... VALUES(uuid(), 'Manuel', 3000000, 'Italy', {'Eurobeat'});
cqlsh:spotify_0802> SELECT * FROM musician_data;

artist_id | artist_name | followers | genres | origin
-----|-----|-----|-----|-----
b5ee5e90-b989-4396-b526-12a661274ffe | Dave Rodgers | 5000000 | {'Dance', 'Eurobeat'} | Italy
e93fc15a-4a80-4dd6-9083-e71d8c59dbdd | YOASOBI | 25000000 | {'J-Pop', 'Pop'} | Japan
1ab26654-2054-437c-97a8-208919485768 | Manuel | 3000000 | {'Eurobeat'} | Italy

(3 rows)
cqlsh:spotify_0802>
```

2. Apa keuntungan utama menggunakan tipe data set pada kolom genres di aplikasi Spotifine dibandingkan jika ia menggunakan tabel relasional (RDBMS) biasa?
 - a. Performa Lebih Cepat: Menghindari operasi JOIN yang berat karena semua genre tersimpan langsung di dalam satu baris kolom.
 - b. Hemat Ruang Penyimpanan: Data disimpan lebih ringkas (biasanya sebagai biner/bitmask) tanpa perlu membuat banyak baris baru secara berulang.
 - c. Skema Lebih Sederhana: Tidak perlu membuat tabel jembatan (junction table) tambahan untuk menghubungkan relasi many-to-many.